

Austin Robinson

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EDUCATION

University of North Carolina at Chapel Hill

May 2026

B.S. Computer Science

Chapel Hill, NC

- **Relevant Coursework:** Data Structures, Algorithms & Analysis, Intro to Probability & Statistics, Data Science & Econometrics, **Artificial Intelligence**, System Security, and Foundations of Software Engineering

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, TypeScript/JavaScript, C, R
- **Frameworks/Tools:** Next.js, React.js, Angular, Node.js, Flask, Clerk, Tailwind, **Docker**, Kubernetes, AWS (EC2), **RESTful API design**, Maven, **Spring Boot**, scalable application architecture
- **Data & Analysis:** SQL, PostgreSQL, SQLAlchemy, Pandas, NumPy, scikit-learn, JSON, Matplotlib
- **AI-Assisted Development:** Uses Claude Code daily as a primary development tool, critically reviewing and iterating on AI-generated code across debugging, prototyping, and workflow automation.

EXPERIENCE

Vechter Home Solutions

2026 – Present

Developer / Sales Representative

Remote

- Built and shipped a full-stack field diagnostics application enabling sales reps to run on-site tests on homeowners' solar and backup power systems and generate automated system data and AI-driven recommendations, in active daily field use.
- Delivered a custom, end-user-facing solution supporting both internal sales staff and field technicians, contributing to at least **\$100K weekly increase** in sales revenue.
- **Own the application end-to-end**, architecting, deploying, and providing ongoing production support for a non-technical user base, including troubleshooting and iteration based on direct field feedback.

ACADEMIC/PERSONAL PROJECTS

LearnWithAI – AI-Powered Lecture Summarizer | Angular, FastAPI, RabbitMQ, Kubernetes

2026

- **Collaborated in a 4-person team** to engineer a modular learning management platform enabling students to upload lecture audio/text and receive automated AI summaries via **Azure OpenAI**.
- **Implemented a real-time event system via WebSockets**, streaming background worker job statuses directly to the frontend client with an automated HTTP polling fallback mechanism.
- **Architected a type-safe backend infrastructure using SQLAlchemy** for data persistence in PostgreSQL, securing endpoints.
- Containerized the application ecosystem using Docker and automated non-mutating validation pipelines (Ruff, Pyright, Prettier) via **GitHub Actions CI/CD**, deploying the final infrastructure onto an OKD Kubernetes cluster.

Extro – Stock Market Forecaster | Next.js, TypeScript, Tailwind, Clerk, Claude API, Vercel

June 2025 – Present

- Developed a full-stack financial dashboard where users log in and simulate long-term portfolio growth using fictional balances, deployed on **Vercel**.
- Integrated the Claude API with structured outputs and prompt caching to generate personalized, ticker-specific investment recommendations from live market data.
- Implemented authentication and per-route authorization with Clerk, and edge-cached market data endpoints to stay within third-party API rate limits at scale.
- Rebuilt the UI on **shadcn/ui** and **Tailwind**, and validated all API responses end-to-end to eliminate malformed-data crashes in the charting layer.

Formal Verification of Client-Side Anti-Cheat System | C++, Alloy, SAT Solvers

February – May 2026

- Conducted a formal security analysis of the *Ultimate AntiCheat* open-source framework by **abstracting C++ source code** into a relational state-transition model.
- Defined critical security properties using First-Order Logic and LTL to **verify system integrity against memory** modification, debugger attachment, and unauthorized module injection.
- Utilized the **Alloy Analyzer** to perform automated **SAT-based model checking**, exhaustively exploring state spaces to identify logic flaws and potential memory "leakage" that could bypass detection.
- **Identified and documented counterexamples** where legitimate system states could be spoofed or where detection mechanisms failed to trigger enforcement, providing **mathematically grounded security assurances**.